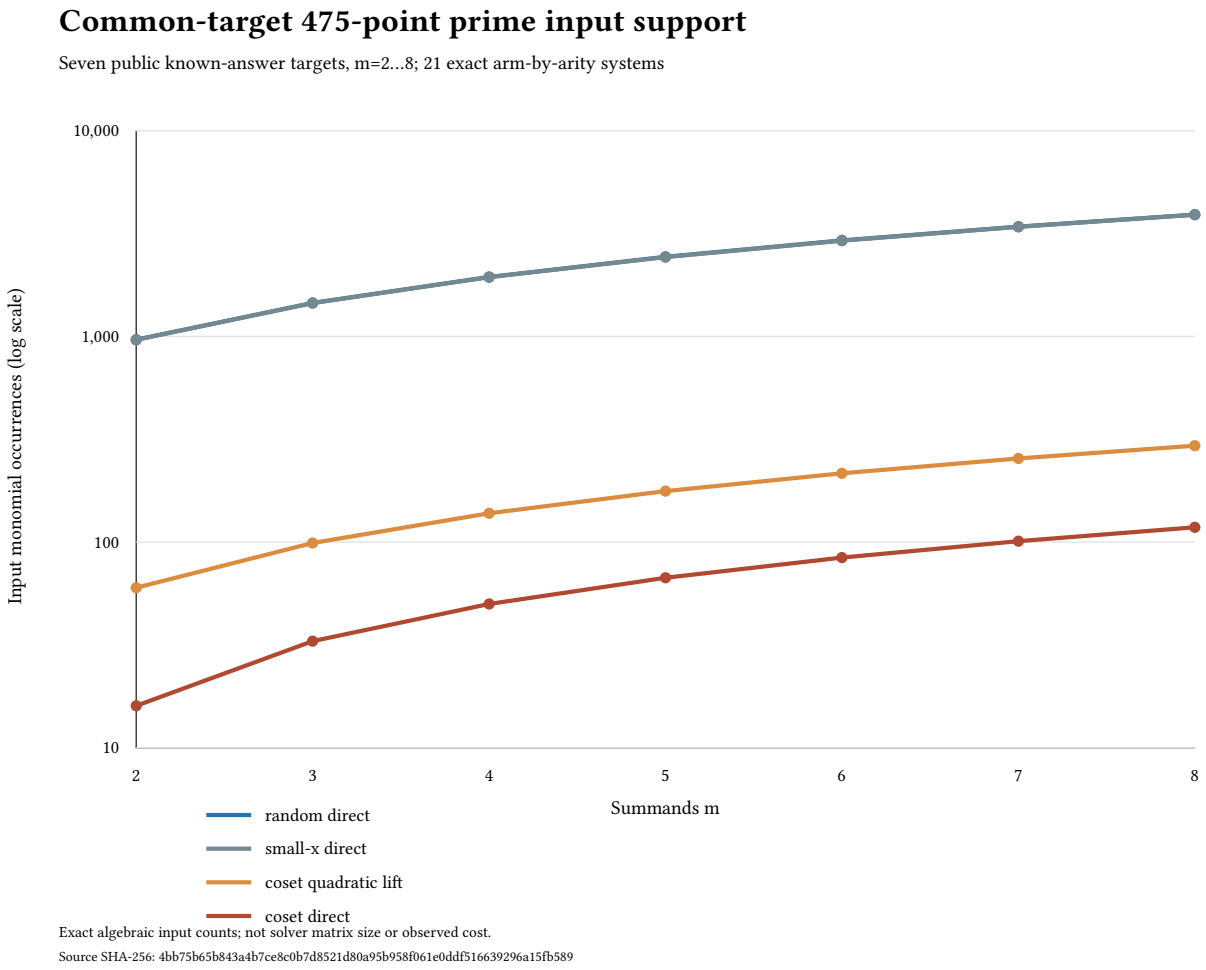
Common target inputs for three 475-point prime bases

Exact public known-answer input preparation and independent Sage replay, 2026-10-17 UTC. Seven targets were derived from the certified 63-bit prime-order curve fixture independently of the three factor bases. The full record serializes 21 chained-S3 systems for summand counts `m=2…8`, plus the coset's exact quadratic membership lifts. Target scalars are public correctness data. There was no solver, relation, regularity or end-to-end measurement.



Graph: input monomial occurrences, log scale. Every arm has 475 actual curve points. At `m=8`, direct coset/random/small-x systems have 118/3910/3902 input occurrences; the coset quadratic lift has 294 occurrences, 103 variables and 104 equations. These counts do not predict solver work. The coset equation also accepts 421 field x-values that do not lift to curve points.

Verification and remaining measurement

Sage independently checked seven target scalar multiplications, all 21 direct membership and S3 systems, and all coset lift gates against the immutable record (SHA-256 `4bb75b65b843a4b7ce8c0b7d8521d80a95b958f061e0ddf516639296a15fb589`). Two initial checker launches were blocked by Sage's cache permissions; the unchanged inputs passed when the runtime could use its cache.

A Coordinator-approved bounded solver protocol is needed to measure solving degree, time and memory, certified point relations, useful rank, linear algebra, a separate unseen same-point target and matched rho, with all setup and failed attempts charged. The installed msolve's 16-bit safety guard prevents a valid direct run at this prime. Direct S8 has seven summands plus a target; the eight-summand row here uses chained S3.

The companion Markdown report links exact inputs, checker, replay and the open full-width measurement problem. The canonical ECDLP frontier graph was checked and needs no update from algebraic input counts.